Mathmatical Analysis Zorich Comprehensions

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Contents

Figure 1: One week, one Zorich

1 Chapter2. The Real Numbers

1.1 Comprehensions

The main content covers the basic properties of real numbers, the definitions and properties of rational and irrational numbers, natural numbers and mathematical induction, and, most importantly, several equivalence principles describing the completeness of real numbers (more generally, the density of real numbers). (Zorich calls these fundamental lemmas, which serve as the foundation for constructing real number theory, but calling them principles is fine.) Finally, some information on Cantor's set theory is added.

The method of describing the completeness of real numbers using Didekind partitioning, which we learned in Fehkingoltz's "Calculus Tutorial (I)" can also be incorporated into this.

Basic Lemmas of the Real Number Completeness Theorem

1. (The least upper (lower) bound principle) Every non-empty set of real numbers that is bounded from above (below) has a unique upper (lower) bound.

2. (Cauchy-Cantor Principle) (The nested interval lemma) The set of closed intervals on the real number axis whose length $\to 0$ determines a unique point (a common point that belongs to all these closed intervals).

3. (Borel-Lebesgue Principle) (The finite covering lemma) Every system of open intervals covering a closed interval contains a finite subcovering.

4. (The Bolzano-Weierstrass Principle) (The limit point lemma) Every bounded infinite set of real numbers has at least one limit point.

Notes. 1. If the length of the closed intervals is not approaching 0, there are still existing common points, but they are not unique.
2. If infinite is limit point, then the Bolzano-Weierstrass Principle is suit for unbounded infinite set of real numbers.

2 important properties of natural numbers $\mathbb{Q}$.

2 Important Properties of Natural Numbers

Thm1. (The Foundamental Theorem of Arithmetic) $\forall n \in \mathbb{N}$, it could be represented uniquely as: $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where p_j are different prime numbers, α_j are positive integers.

Thm2. (The Principle of Archimedes) Given $h > 0$, then $\forall x \in \mathbb{R}$, $\exists$ unique $k \in \mathbb{Z}$, s.t. $(k-1)h \leq x \leq kh$.

Cantor Set Theory.

Countable and Uncountable Sets

Proposition 1. card(Countable Set)=card($\mathbb{N}$)=$\aleph_0$, card(Uncountable Set)=card($\mathbb{R}$)=$\aleph_1$ (cardinality of the continuum).

Proposition 2. Any subset of a at most countable set (card(X)$\leq$card($\mathbb{N}$)) is at most countable; the at most countable union of at most countable sets is at most countable.

Proposition 3. (Corollary) The direct sum (or direct product) of countable sets is countable. (Corollary) The set of natural numbers or algebraic numbers is countable.

Thm 1. (Cantor's Theorem) card($\mathbb{N}$) $<$ card($\mathbb{R}$).

Proposition 4. (Corollary) the irrational numbers or transcendental numbers exisst, and the set of all of them is uncountable.

1.2 Problems Sets

An interesting example-the Cantor set

Example. Let $x \in [0,1] \subset \mathbb{R}$, its ternary expansion is $x = 0.a_1a_2a_3\cdots$ ($a_i \in \{0,1,2\}$) satisfying: $\forall i \in \mathbb{N}, a \neq 1$, then the set of all such elements is called the Cantor set.

Notes. The Cantor set is uncountable (It's easy: the elements of it can be onto mapped to all binary decimals on $[0,1]$), and we'll then learn: Cantor set is measure zero in sense of Lebesgue.

2 2 Classical Limits

2.1 Original

2.1.1 $\lim_{n\to\infty} \frac{a^n}{n^k}$

The problem is:

$$\lim_{n\to\infty} \frac{a^n}{n^k} \quad \text{where } a > 1,\ k > 0$$

Solution (Solution1). Based on **Binomial Theorem**
$a^n = [a \triangleq 1 + \lambda] = 1 + n\lambda + \frac{n(n-1)}{2}\lambda^2 + \cdots > \frac{n(n-1)}{2}\lambda^2$
also $n - 1 > \frac{n}{2},\ as\ n > 2$, then $a^n > \frac{n^2}{4}(a-1)^2$.
As $k = 1$, $\lim_{n\to\infty} \frac{a^n}{n} = \infty$, and as $k > 1$, $\lim_{n\to\infty} \frac{a^n}{n^k} = \lim_{n\to\infty} (\frac{(a^{\frac{1}{k}})^n}{n})^k = \infty$.

Solution (Solution2). Based on **L'Hospital's Rule**
$\lim_{n\to\infty} \frac{a^n}{n^k} = \lim_{n\to\infty} \frac{\ln a \cdot a^n}{k \cdot n^{k-1}} = \lim_{n\to\infty} \frac{(\ln a)^2 \cdot a^n}{k(k-1) \cdot n^{k-2}} = \cdots = \lim_{n\to\infty} \frac{(\ln a)^k \cdot a^n}{k!} = \infty$.

Solution (Solution3). (Haven't found a totally different solution yet.)

2.1.2 $\lim_{n\to\infty} n^{\frac{1}{n}}$

The problem is:

$$\lim_{n\to\infty} n^{\frac{1}{n}}$$

Solution (Solution1). Due to 1. $a^n > \frac{n^2}{4}(a-1)^2$, let $a = n^{\frac{1}{n}}$, then $0 < n^{\frac{1}{n}} < \frac{2}{\sqrt{n}}$. Thus $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

Solution (Solution2). Based on Mean Value Inequalities Chain(GM-AM): $n^{\frac{1}{n}} = (\sqrt{n} \cdot \sqrt{n} \cdot 1 \cdots 1)^{\frac{1}{n}} \leq \frac{2\sqrt{n}+n-2}{n}$, then $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$.

2.2 Some repurposed variants

2.2.1 $\lim_{n\to+\infty} \sqrt[n]{n!}$

The problem is:

$$\lim_{n\to+\infty} \sqrt[n]{n!}$$

Solution (Solution1). Based on **Stirling's Approximation**: $n! \sim \sqrt{2\pi n}(\frac{n}{e})^n$, then $\sqrt[n]{n!} \sim \sqrt[n]{\sqrt{2\pi n}}(\frac{n}{e}) = \frac{n}{e}\cdot(2\pi n)^{\frac{1}{2n}}$. Thus $\lim_{n\to+\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution2). $n! \geq \left(\frac{n}{2}\right)^{\frac{n}{2}}$ because half of the factors are at least $\frac{n}{2}$

Solution (Solution3). Consider Taylor series, $e^x \geq \frac{x^n}{n!}$, $\forall x \geq 0, n \in \mathbb{N}$, then $n! \geq \frac{x^n}{e^x}$, take $x = n$ problem solved.

Solution (Solution4). Based on **Stolze Cesaro Theorem**: $\lim\limits_{n\to+\infty} \sqrt[n]{n!} = \lim\limits_{n\to+\infty} \frac{n!^{\frac{1}{n}}}{1} = \lim\limits_{n\to+\infty} \frac{\ln(n!)}{n} = \lim_{n\to+\infty} \frac{\ln(n)+\ln(n-1)+\cdots+\ln(1)}{n}$, then apply Stolze Cessaro Theorem, $\lim\limits_{n\to+\infty} \frac{\ln(n)+\ln(n-1)+\cdots+\ln(1)}{n} = \lim\limits_{n\to+\infty} \frac{\ln(n+1)}{1} =$, then $\lim\limits_{n\to+\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution5). Using a multiplicative variant of Gauss' trick we get: $(n!)^2 = (1\cdot n)(2\cdot(n-1))\cdots((n-1)\cdot 2)(n\cdot 1) \geq n^n$.

3 Chapter 3.Limits

3.1 Comprehensions

3.1.1 ξ.1 The Limit of a Sequence

First, the most basic is the ε-N definition of sequence limits and the basic properties of limits (existence $\implies$ boundedness). On this basis, the operational processes of limits are established (four arithmetic operations and inequality relations, note: the strictness of inequalities may not be maintained after the limit process). Then, we proceed to the discussion of the existence of limits (introducing two types of convergence criteria: the /general/ Cauchy convergence criterion and the /special - monotonic sequence/ Weierstrass theorem), and an important method: subsequences, subsequence limits, and upper and lower limits.

Furthermore, this section also introduces some preliminary knowledge of series. The essential idea is to treat series as partial sum sequences, so that the theory of sequence limits above can be directly applied to obtain convergence tests for series (also divided into general and special two types) and introduces absolute convergence as a stronger form of convergence (unaffected by rearrangement), and the related properties of convergent series (necessary condition - terms $\to 0$). On this basis (mainly the Weierstrass theorem), some useful corollaries are made, such as: the comparison test (only applicable when all terms are non-negative or all terms have the same sign), the Cauchy convergence criterion (based on the relationship between $\sup\lim_{n\to\infty} |a_n|^{1/n}$ and 1) and the d'Alembert test (based on the relationship between $\sup\lim_{n\to\infty} |\frac{a_{n+1}}{a_n}|$ and 1) (applicable in general cases and ensures absolute convergence), and the Cauchy condensation test ($\sum_{n=1}^{\infty} a_n$ and $\sum_{k=0}^{\infty} 2^k a_{2^k}$ have the same convergence behavior) (applicable when terms are non-negative and decreasing).

Examples

1. For the series $\sum_{n=1}^{\infty}(2+(-1)^n)^n x^n$, it converges for $|x| < \frac{1}{3}$ and diverges for $|x| > \frac{1}{3}$ and $|x| = \frac{1}{3}$, using the Cauchy root test.
2. For the series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$, it converges for all $x \in \mathbb{R}$, using the d'Alembert ratio test. Note: This can also be judged by the Cauchy root test, based on $\lim_{n\to+\infty} \sqrt[n]{n!} = +\infty$.
3. For the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, it converges for $p > 1$ and diverges for $p \leq 1$, using the Cauchy condensation test.

3.1.2 ξ.2 The Limit of a Function

The study of function limits largely follows the methods of sequence limits. First, the ε-δ definition of a function limit is introduced (note that function limits are defined on a punctured neighborhood), along with the properties of a function's limit existing at a point (existence $\implies$ local boundedness). On this basis, the four arithmetic operations of function limits are studied (the proof process will become much more convenient if an infinitesimal quantity is first introduced as an equivalent definition of the limit of function) and the inequality relations during the limit process (note: the strictness of inequalities may not be maintained after the limit process). Furthermore, the limit behavior of elementary functions and the limit behavior of composite functions are explained (here, it is necessary to learn two important limits).

A novel aspect of Zorich's approach is his definition of something called a "base" (essentially, a family of sets satisfying certain conditions, which facilitates the description of "approaching" in the limit process); he then re-describes the aforementioned content based on this concept. Next, the discussion proceeds to the problem of the existence of limits (again introducing two methods: the /general/ Cauchy convergence criterion and the /special - monotonic function/ Weierstrass theorem). Finally, the issue of asymptotic comparison of functions (order estimation) is briefly introduced.

2 Important Limits

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1$.
2. $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x = e$.

3.2 Problems Sets

3.2.1 ξ.1 The Limit of a Sequence

Problem 1. (C3S1Q3)
We mark all the points on a circle obtained from a fixed point by rotations of the circle through angles of n radians, where $n \in \mathbb{Z}$ ranges over all integers. Describe all the limit points of the set so constructed.

Solution. Attempt 1. $n(mod\ \alpha) = n - \alpha \left\lfloor \frac{n}{\alpha} \right\rfloor$, $\forall \alpha \in \mathbb{R}$.
Attempt 2. $\forall \varepsilon > 0$, by pigeonhole principle ($\forall N > 0, as\ n \to \infty$, $n - N$ of integers ($n - N \to \infty$) will be located between those N integers.) $\exists m, n < \frac{1}{\varepsilon}$, s.t. $|m - n|(mod\ 2\pi) < \varepsilon$,
then let $k = |m - n|$, (construct full proof for dense property), $\forall x \in [0, 2\pi]$, $x \in \left[k \left\lfloor \frac{x}{k(mod\ 2\pi)} \right\rfloor, k\lceil \frac{x}{k(mod\ 2\pi)} \rceil\right]$ and the length of interval equals to $k < \varepsilon$, which means $\mathbb{R}_{2\pi}$ is dense in $(0, 2\pi)$ or $[0, 2\pi]$.

Development:
Kronecker's approximation theorem.(Wolfram MathWorld, Wikipedia)

(Variants)
Prove: $\{n^p \pmod \pi) : p > 0\}$ is dense on $[0, \pi]$.

Proof. Core: Similar to the idea of differential (tool: difference), reduce the order, and finally turn it into the prototype of the above problem.
Let $p > 0$ be an arbitrary real number. We claim that $\{n^p \pmod \pi) : p > 0\}$ is dense on $[0, \pi]$.
For any function $f(n)$ let $\Delta f(n) = f(n+1) - f(n)$. If $f(n)$ converges to 0 $\pmod \pi$ then so does $\Delta f(n)$. Now for $f(n) = n^p$ we have $\Delta f(n) \approx pn^{p-1}$, $\Delta^2 f(n) \approx p(p-1)n^{p-2}$, etc. More precisely, by Newton's binomial theorem, $(n+1)^p - n^p = \sum_{i=1}^{\infty} \binom{p}{i} n^{p-i}$.
Let $k = \lfloor p \rfloor$. One can check that the part of the series with $i > k$ has the overall form $o(1)$. Thus $\Delta(n^p) = pn^{p-1} + c_2 n^{p-2} + \cdots + c_k n^{p-k} + o(1)$ for various constant coefficients $c_2, \ldots, c_k$. Iterating, it follows that $\Delta^k(n^p) = (p)_k n^{p-k} + o(1)$, where $(p)_k = p(p-1)\cdots(p-k+1)$. Thus the function $g(n) = (p)_k n^{p-k}$ tends to 0 mod π, but this is impossible: since π is irrational and we must have $p > k$, but then $g(n) \to \infty$ while $\Delta g(n) \to 0$ (There aren't $\pmod \pi$), which means $g(n)$ tends to ∞ "densely".), so the range of g is dense mod π.

Problem 2. (C3S1Q5)

a) $1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n} = 3 - \frac{1}{1 \cdot 2 \cdot 2!} - \cdots - \frac{1}{(n-1) \cdot n \cdot n!}$ holds, as $n \geq 2$.

b) $e = 3 - \sum_{n=0}^{\infty} \frac{1}{(n+1) \cdot (n+2) \cdot (n+2)!}$.

c) For computing e,$e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} + \frac{1}{n!n}$ is much better than $e \approx 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$.

Quick Solution. a) $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}+\frac{1}{1\cdot2\cdot2!}+\cdots+\frac{1}{(n-1)\cdot n\cdot n!}$ $=\left[\frac{1}{n!}+\frac{1}{n!n}+\frac{1}{(n-1)n\cdot n!}-\frac{1}{(n-1)\cdot(n-1)!}\right]=1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{(n-1)!}+\frac{1}{(n-1)\cdot(n-1)!}+\frac{1}{1\cdot2\cdot2!}+\cdots+\frac{1}{(n-2)\cdot(n-1)\cdot(n-1)!}=\cdots=1+\frac{1}{1!}+\frac{1}{1\cdot1!}=3$

b) $\lim_{n\to\infty}1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}$ and $\lim_{n\to\infty}1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}$, $\Delta_n=\frac{1}{n!n}=o(1), n\to\infty$.

c) Since both $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}+\frac{1}{n!n}$ and $1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}$ are monotonically increasing $\to e$, and the former has an extra term $\frac{1}{n!n}$ thus it grows faster (i.e., converges faster). To estimate the difference in convergence speed, we just need to find the maximum $k\in\mathbb{N}^+$ such that $\frac{1}{n!n}\geq\frac{1}{(n+1)!}+\cdots+\frac{1}{(n+k)!}$.
(Actually, $\forall n\geq 2, k\in\mathbb{N}^+$ i.e $Max k=+\infty$)

Problem 3. (C3S1Q6)
Given$S_p(a,b)=\left(\frac{a^p+b^p}{2}\right)^{\frac{1}{p}}, p\neq 0, a,b>0$, prove:

a) $\forall p\neq 0, a,b>0, S_p(a,b)\in[a,b]$;

b) $\lim_{n\to\infty}S_n(a,b)=b, \lim_{n\to\infty}S_{-n}(a,b)=a$.

Quick Solution. a) It is sufficient to define the order relation of a and b, for example: without loss of generality, assume $a<b$.

b) It is sufficient to define the relative size relationship of a and b (i.e., what is commonly called substitution or elimination by increment), for example: without loss of generality, let $b=\lambda a$.

Thoughts:
(Trick) When a problem has multiple (≥ 2) variables, it may be helpful to look for connections between these seemingly "independent" variables (for example, in question 6.b) of this section).

Problem 4. (C3S1Q7) (Classical problem)
If $a>0$, then sequence $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)$ converges to $\sqrt{a}$.

Quick Solution. $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)\geq\sqrt{a},\ n\geq 1$, i.e. $\forall x_1>0,\ x_2\geq\sqrt{a}$.
[Induction] then $\{x_n\}_{n=2}^{\infty}$ monotonically decrease, $\lim_{n\to\infty}x_{n+1}=\lim_{n\to\infty}\frac{1}{2}\left(x_n+\frac{1}{x_a}\right)=\sqrt{a}$.

Thoughts:
(Experience) An experienced student, upon seeing a recurrence relation, can immediately guess that the problem will most likely require the use of methods like the Weierstrass theorem to first determine convergence, and then take the limit on both sides of the equation.

(Tip) If starting from a general problem-solving approach, the first step is usually to determine the basic properties of the sequence (positivity/negativity, monotonicity, parity, periodicity (or these properties starting from a certain term), etc.), and then consider which method to use to address convergence (general $\to$ Cauchy convergence criterion, monotonic $\to$ Weierstrass theorem) and other issues.

(Follow-up Question)
Please estimate the convergence rate, i.e., the dependence of $x_n - \sqrt{a}$ on n.

Problem 5. (C3S1Q8)

a) (Faulhaber's formula) $S_k(n) = \sum_{i=1}^{n} i^k = \frac{1}{k+1}\sum_{r=0}^{k} C_{k+1}^r B_r n^{k+1-r}$, B_r is Bernoulli number.

b) $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \frac{1}{k+1}$.

Solution. a) This part is a little hard, because to wholy solving it, we needs expotential generating function with indeterminate z: $G(z,n) = \sum_{k=0}^{\infty} S_k(n)\frac{1}{k!}z^k$. You could get it from wikipedia.

b) Method1. $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \frac{i^k}{n^k} = \int_0^1 x^k dx = \frac{1}{k+1}$ (Riemann sum)

Method2. $\lim_{n\to\infty} \frac{S_k(n)}{n^{k+1}} = \lim_{n\to\infty} \frac{(n+1)^k}{(n+1)^{k+1}-n^{k+1}}$ (Stolze-Cesaro Lemma) $= \lim_{n\to\infty} \frac{1}{(n+1)-\left(\frac{n}{n+1}\right)^k \cdot n} = \lim_{n\to\infty} \frac{1}{1+n\left[1-\left(\frac{n}{n+1}\right)^k\right]}$ (order estimation) $= \lim_{n\to\infty} \frac{1}{1+n\left[1-\left(1-k\cdot\frac{1}{n+1}+O\left(\frac{1}{n^2}\right)\right)\right]} = \lim_{n\to\infty} \frac{1}{1+\frac{n}{n+1}\cdot k+O\left(\frac{1}{n}\right)} = \frac{1}{1+k}$.

3.2.2 ξ.2 The Limit of a Function

Problem 6. (C3S2Q1)
Porve that:

a) there $\exists$ a unique function defined on $\mathbb{R}$ satisfying : $f(1) = a$ (where $a > 0, a \neq 1$), $f(x_1)\cdot f(x_2) = f(x_1 + x_2)$, $\lim_{x\to x_0} f(x) = f(x_0)$.

b) there $\exists$ a unique function defined on $\mathbb{R}^+$ satisfying : $f(a) = 1$ (where $a > 0, a \neq 1$), $f(x_1) + f(x_2) = f(x_1 \cdot x_2)$, $\lim_{x\to x_0} f(x) = f(x_0)$.

Quick Solution. By substitution (for example: $x \triangleq e^{\xi}$, etc.), transform the given variant of the Cauchy equation back into the form $f(x+y) = f(x)+f(y)$ (uniqueness is obtained from the equation itself and continuity, and the method has been provided in the discussion of the corresponding equation in Fikhtengoltz's "Calculus Tutorial (I)").

Problem 7. (C3S2Q2)

a) Provide an isomorphism mapping between the group $(\mathbb{R}, +)$ and the group $(\mathbb{R}^+, \cdot)$;

b) Prove that $(\mathbb{R}, +)$ and the group $(\mathbb{R} \setminus 0, \cdot)$ are not isomorphic.

> *Quick Solution.* a) $\forall x, y \in \mathbb{R}$, if $\varphi(x+y) = \varphi(x) \cdot \varphi(y)$, then $\varphi(x) = e^{cx}$;
>
> b) From the fact that "isomorphism not only is a bijection but also preserves the group operation", if $\varphi : \mathbb{R} \to \mathbb{R}\backslash 0$ exists, then from $\varphi(0) = 1$, it follows that $\varphi(x) \cdot \varphi(-x) = \varphi(0) = 1$.

Problem 8. (C3S2Q3)

> *Quick Solution.* a) $\lim\limits_{x\to+0} x^x = \lim\limits_{x\to+0} e^{x\ln(x)} = \lim\limits_{t\to+\infty} e^{-\frac{\ln(t)}{t}} = 1$;
>
> b) $\lim\limits_{x\to+\infty} x^{\frac{1}{x}} = \lim\limits_{x\to+\infty} e^{\frac{\ln(x)}{x}} = 1$;
>
> c) $\lim\limits_{x\to 0} \frac{\log_a(1+x)}{x} = \lim\limits_{x\to 0} \frac{1}{\ln a} \cdot \frac{\ln(1+x)}{x} = \left[\ln(1+x) = x - \frac{1}{2}x^2 + O(x^3)\right] = \frac{1}{\ln a}$;
>
> d) $\lim\limits_{x\to 0} \frac{a^x - 1}{x} = [a^x - 1 \triangleq t] = \lim\limits_{x\to 0} \frac{t}{\log_a(1+t)} = \ln a$.
>
> **Notes.** Both a) and b) used $\lim_{x\to+\infty} \frac{\log_a x}{x^\alpha} = 0,\ \alpha > 0$.

Problem 9. (C3S2Q4)
Prove: $1 + \frac{1}{2} + \cdots + \frac{1}{n} = \ln n + c + o(1),\ n \to \infty$.

> *Quick Solution.* Using $\ln(n+1) - \ln n = \ln \frac{n+1}{n} = \ln\left(1 + \frac{1}{n}\right) = \frac{1}{n} + O\left(\frac{1}{n^2}\right),\ n \to \infty$, then Σ them.

Problem 10. (C3S2Q5)

a) If two positive term series $\sum_{n=1}^\infty a_n$ and $\sum_{n=1}^\infty b_n$ satisfy: $a_n \sim b_n, n \to \infty$, then these two series have the same convergence behavior;

b) The series $\sum_{n=1}^\infty \sin \frac{1}{n^p}$ converges only when $p > 1$.

Solution. a) $a_n \sim b_n$ i.e. $\forall \varepsilon > 0, \exists N > 0, s.t. \forall n > N, |\frac{a_n}{b_n} - 1| < \varepsilon$ i.e. $1-\varepsilon < \frac{a_n}{b_n} < 1+\varepsilon, \forall n > N$, take a specific $\varepsilon = \varepsilon_0$, then $(1-\varepsilon_0)b_n < a_n < (1+\varepsilon_0)b_n$. Convergence/divergence is thereby controlled.

b) $p > 0 : \sin \frac{1}{n^p} \sim \frac{1}{n^p}$ Two positive term series have the same convergence behavior, the convergence/divergence of the latter can be given by the Cauchy condensation test (see Summary C3S1 for details)
$p = 0 : \sin \frac{1}{n^p} = \sin 1 > 0$ diverges
$p < 0 : \sum_{n=1}^{\infty} \sin \frac{1}{n^p} = [q \triangleq -p > 0] = \sum_{n=1}^{\infty} \sin n^q$ diverges (Essentially, $\{n^q (\bmod \pi)\}$ is dense on $[0, \pi]$, details in (C3S1Q3 variants).

Problem 11. (C3S2Q6)

a) If $\forall n \in \mathbb{N}$ we have $a_n \geq a_{n+1} > 0$, and the series $\sum_{n=1}^{\infty} a_n$ converges, then $a_n = o\left(\frac{1}{n}\right)$, $n \to \infty$;

b) If $b_n = o\left(\frac{1}{n}\right)$, then it is always possible to construct a convergent series $\sum_{n=1}^{\infty} a_n$; such that $b_n = o(a_n)$, $n \to \infty$;

c) If the positive term series $\sum_{n=1}^{\infty} a_n$ converges, then the series $\sum_{n=1}^{\infty} A_n$, as $A_n = \sqrt{\sum_{k=n}^{\infty} a_k} - \sqrt{\sum_{k=n+1}^{\infty} a_k}$, also converges, and $a_n = o(A_n), n \to \infty$;

d) If the positive term series $\sum_{n=1}^{\infty} a_n$ diverges, then the series $\sum_{n=1}^{\infty} A_n$, as $A_n = \sqrt{\sum_{k=1}^{n} a_k} - \sqrt{\sum_{k=1}^{n-1} a_k}$, also diverges, and $A_n = o(a_n), n \to \infty$.

Solution. a) Applying the Cauchy convergence criterion to the series $\sum_{n=1}^{\infty} a_n$, we have: $\forall \varepsilon > 0, \exists N > 0$ such that $\forall n, m > N, |a_n + \cdots + a_m| < \varepsilon$. If we set $m = 2n$ and since a_n is monotonically decreasing, then $n a_{2n} < \varepsilon$, i.e. $a_n = o\left(\frac{1}{n}\right)$;

b) Consider alternating series and monotonic sequences: $a_n \triangleq (-1)^n \sup_{k \geq n} b_k$, then $\sum_{n=1}^{\infty} a_n$ converges. (Leibniz Theorem) Or just let $a_n = (-1)^n \frac{1}{n}$.

c) d) Easy to prove. (Cancellation of intermediate terms in summation; rationalization, division).

Thoughts:
From c) and d), it can be seen: there always exists a convergent positive term series that is "larger" than a given convergent positive term series, and there always exists a positive term series that is "smaller" than a given divergent positive term series.
There is no unique, universal standard series that can be applied to all comparison tests.

Problem 12. (C3S2Q7&Q8)

a) The series $\sum_{n=1}^{\infty} \ln a_n, a_n > 0, n \in \mathbb{N}$ converges i.f.f. the sequence $\{\Pi_n = a_1 \cdots a_n\}$ has a nonzero limit;

b) If the infinite product $\prod_{n=1}^{\infty} e_n$ converges, then $\lim_{n\to\infty} e_n = 1$;

c) The series $\sum_{n=1}^{\infty} \ln(1+a_n), |a_n| < 1$ converges absolutely i.f.f. the series $\sum_{n=1}^{\infty} a_n$ converges absolutely;

d) The infinite product $\prod_{n=1}^{\infty} e_n, e_n > 0$ converges i.f.f. the series $\sum_{n=1}^{\infty} \ln e_n$ converges;

e) If $e_n = 1 + a_n$ and a_n all have the same sign, then the infinite product $\prod_{n=1}^{\infty}(1+a_n)$ converges i.f.f. the series $\sum_{n=1}^{\infty} a_n$ converges.

Quick Solution. a) Easy to prove. $\sum_{n=1}^{\infty} \ln a_n = \ln \prod_{n=1}^{\infty} a_n$ (they have the same convergence behavior).

b) Easy to prove. (If $\prod_{n=1}^{\infty} e_n = c \neq 0$, then $\lim_{n\to\infty} e_n = \lim_{n\to\infty} \frac{\prod_{k=1}^{n} e_k}{\prod_{k=1}^{n-1} e_k} = 1$. Alternatively, take logarithms and apply the basic properties of series convergence.)

c) If $\sum_{n=1}^{\infty} \ln(1+a_n)$ converges, then by a) $\prod_{n=1}^{\infty}(1+a_n)$ has a nonzero limit. Further, by b), $a_n \to 0$ as $n \to \infty$, and since $|\ln(1+a_n)| \sim \ln(1+|a_n|) \sim |a_n|$, it follows (from C3S2Q5 a)) that the statement is proved.

d) Easy to prove. (Same as a))

e) $\ln(1+a_n) \sim a_n$, so $\prod_{n=1}^{\infty}(1+a_n)$ converges if and only if $\sum_{n=1}^{\infty} \ln(1+a_n)$ converges if and only if $\sum_{n=1}^{\infty} a_n$ converges.

Problem 13. (C3S2Q9)

a) Calculate $\Pi_{n=1}^{\infty}(1+x^{2^{n-1}})$;

b) Caculate $\Pi_{n=1}^{\infty} \cos\left(\frac{x}{2^n}\right)$, and prove $\frac{\pi}{2} = \frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\cdot\sqrt{\frac{1}{2}}}}\cdots}$ (Vieta's formula);

c) Find the $f(x)$, s.t. $f(0) = 1$, $f(2x) = \cos^2 x \cdot f(x)$, $\lim_{x\to 0} f(x) = f(0)$.

Solution. a) From C3S2Q7 e), $\prod_{n=1}^{\infty}(1+x^{2^n-1})$ and $\sum_{n=1}^{\infty} x^{2^n-1}$ have the same convergence behavior, i.e., they converge only when $|x| < 1$, and $\prod_{n=1}^{\infty}(1+x^{2^n-1}) = \frac{1-x^{2^n}}{1-x}$ (generalized power series multiplication theorem);

b) $\prod_{n=1}^{\infty} \cos\left(\frac{x}{2^n}\right) = \lim_{n\to\infty} \prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) = \lim_{n\to\infty} \frac{\sin\frac{x}{2^n}\prod_{k=1}^{n}\cos\frac{x}{2^k}}{\sin\frac{x}{2^k}} = \lim_{n\to\infty} \frac{\sin x}{2^n \sin\frac{x}{2^n}} = \lim_{n\to\infty} \frac{\sin x}{x}\cdot\frac{\frac{x}{2^n}}{\sin\frac{x}{2^n}} = \frac{\sin x}{x}$ Due to $\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$ and (above) $\prod_{n=1}^{\infty}\cos\frac{x}{2^n} = \frac{\sin x}{x}$ (let $x = \frac{\pi}{2}$), Vieta's formula is proved;

c) Using the recurrence relation $f(2x) = \cos^2 x \cdot f(x)$ and taking $\frac{x}{2^n}, n = 0, 1, 2, \ldots$ yields, $f(x) = \left(\frac{\sin x}{x}\right)^2$.

Problem 14. (C3S2Q10)

a) If $\frac{b_n}{b_{n+1}} = 1+\beta_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \beta_n$ converges absolutely, then the limit $\lim_{n\to\infty} b_n = b$ evidently exists;

b) If $\frac{a_n}{a_{n+1}} = 1 + \frac{p}{n} + \alpha_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \alpha_n$ converges absolutely, then $a_n \sim \frac{c}{n^p}, n \to \infty$;

c) If $\frac{a_n}{a_{n+1}} = 1 + \frac{p}{n} + \alpha_n, n = 1, 2, \cdots$, and the series $\sum_{n=1}^{\infty} \alpha_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n$ converges when $p > 1$ and diverges when $p \leq 1$ (Gauss's test for absolute convergence of series).

Solution. a) From C3S2Q7 and Q8 c), it follows that $\sum_{n=1}^{\infty} \ln(1 + \beta_n) = \sum_{n=1}^{\infty} \ln\left(\frac{b_n}{b_{n+1}}\right)$ and $\sum_{n=1}^{\infty} \beta_n$ have the same convergence behavior, both absolutely convergent.

b) (Good question, good solution)
Method 1. Consider $\frac{n^p}{(n+1)^p}\frac{a_n}{a_{n+1}} = \frac{n^p}{(n+1)^p}\left(1 + \frac{p}{n} + \alpha_n\right) = \left(1 + \frac{1}{n}\right)^{-p} + \left(1 + \frac{1}{n}\right)^{-p}\frac{p}{n} + \left(1 + \frac{1}{n}\alpha_n\right)^{-p} = \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\left(1 + \frac{p}{n}\right) + \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\alpha_n \left[\triangleq 1 + \theta_n\right]$,
$\theta_n = \left(1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)\right)\left(1 + \frac{p}{n} + \alpha_n\right) - 1 = O\left(\frac{1}{n^2}\right)\left(1 + \frac{p}{n} + \alpha_n\right) - \frac{p^2}{n^2} + \alpha_n\left(1 - \frac{p}{n}\right)$, $|\theta_n| \leq |O\left(\frac{1}{n^2}\right)\left(1 + \frac{p}{n} + \alpha_n\right)| + |\frac{p^2}{n^2}| + |\alpha_n\left(1 - \frac{p}{n}\right)|$, and $\sum_{n=1}^{\infty} \alpha_n$ abs. cgt. , then $\sum_{n=1}^{\infty} \theta_n$ abs. cgt. Then due to C3S2Q7 c), $\sum_{n=1}^{\infty} \ln(1 + \theta_n) = \sum_{n=1}^{\infty} \ln\frac{n^p a_n}{(n+1)^p a_{n+1}} = \ln\alpha_1 - \lim_{n\to\infty} n^p\alpha_n$ abs. cgt. i.e. $n^p\alpha_n \sim c, n \to \infty$. $(\alpha_n \sim \frac{c}{n^p})$.
Method 2 (Optimize). Following the first step of Method 1:
$\ln\frac{n^p a_n}{(n+1)^p a_{n+1}} = \ln\frac{\alpha_n}{\alpha_{n+1}} - p\ln\left(1 + \frac{1}{n}\right) = \ln\left(1 + \frac{p}{n} + \alpha_n\right) - p\ln\left(1 + \frac{1}{n}\right) = \frac{p}{n} + \alpha_n + O\left(\frac{p}{n} + \alpha_n\right)^2 - p\left(\frac{1}{n} + O\left(\frac{1}{n^2}\right)\right) = \alpha_n + O\left(\frac{1}{n^2}\right) + O(\alpha_n) \to$ abs. cgt. going on Method 1.

c) From b) and the convergence behavior of $\sum_{n=1}^{\infty}\frac{1}{n^p}$, it immediately follows.

Problem 15. (C3S2Q11)

Prove: For any sequence $\{a_n\}$, $\limsup\left(\frac{a_{n+1}}{a_n}\right)^n > e$, and this estimate is the best possible.

Solution. (Great deal: Wonderful observation!) /Contradiction/
Lemma: $\forall x \in (0, e)$, for sufficiently large n, $x^{\frac{1}{n}} < 1 + \frac{1}{n}$.
If $\overline{\lim\limits_{n\to\infty}}\left(\frac{1+a_{n+1}}{a_n}\right)^n < e$, $a_n > 0$, then $\exists b$, s.t. $\overline{\lim\limits_{n\to\infty}}\left(\frac{1+a_{n+1}}{a_n}\right)^n < b < e$, i.e. $\exists$ sufficient large n, s.t. $\left(\frac{1+a_{n+1}}{a_n}\right)^n < b < e$, i.e. (using std::Lemma) $\frac{1+a_{n+1}}{a_n} < b^{\frac{1}{n}} < \frac{n+1}{n}$, let $b_n \triangleq \frac{a_n}{n}$, then $b_{n+1} < b_n - \frac{1}{n+1}$, increasing to $n+m$ and Σ them: $b_{n+m} < b_n - \sum_{k=1}^{m}\frac{1}{n+k}$, fixing b_n and letting $m \to \infty$ we have $b_{n+m} = \frac{a_{n+m}}{n} \to -\infty,\ m \to \infty$ which derivate constraction.

4 Basel Problem

4.1 $\zeta(2) = \sum\limits_{n=1}^{\infty} \frac{1}{n^2}$

Problem 16. Calculate (or prove)

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Solution. Method 1. Double Integral.
Consider the double integral $I = \iint_{D_1} \frac{1}{1-xy}\, dx\, dy$, where $D_1 = \{(x,y) : 0 < x < 1, 0 < y < 1\}$.

$$\begin{aligned}\iint_{D_1} \frac{dx\,dy}{1-xy} &= \text{/Generalized power series multiplication theorem/} \\ &= \iint_{D_1} \sum_{n=0}^{\infty} (xy)^n\, dx\, dy = \int_0^1 \left(\int_0^1 \sum_{n=0}^{\infty} (xy)^n\, dx \right) dy \\ &= \int_0^1 \sum_{n=1}^{\infty} \frac{y^{n-1}}{n}\, dy = \sum_{n=1}^{\infty} \frac{1}{n^2}.\end{aligned}$$

/Transformation of coordinates or variable substitution/

$$\begin{aligned}&[x = \frac{u-v}{\sqrt{2}}, y = \frac{u+v}{\sqrt{2}}, \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} x'_u & x'_v \\ y'_u & y'_v \end{vmatrix} \\ &\Leftarrow (u,v)^T = \begin{pmatrix} \cos -\frac{\pi}{4} & -\sin -\frac{\pi}{4} \\ \sin -\frac{\pi}{4} & \cos -\frac{\pi}{4} \end{pmatrix} (x,y)^T]\end{aligned}$$

$$\begin{aligned}I &= \iint_{D_2} \frac{1}{1 - \frac{u^2-v^2}{2}} \cdot 1\, du\, dv \\ &= 2\left[\iint_{D_21} \frac{1}{2-u^2+v^2}\, du\, dv + \iint_{D_22} \frac{1}{2-u^2+v^2}\, du\, dv \right] \\ &= 4\left[\int_0^{\frac{\sqrt{2}}{2}} \left(\int_0^u \frac{dv}{2-u^2+v^2} \right) du + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \left(\int_0^{\sqrt{2}-u} \frac{dv}{2-u^2+v^2} \right) du \right] \\ &= [u \triangleq \sqrt{2}\sin\theta, v \triangleq \sqrt{2}\cos\theta] \\ &= \frac{\pi^2}{18} + \frac{\pi^2}{9} = \frac{\pi^2}{6}.\end{aligned}$$

i.e. $\zeta(2) = \sum\limits_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

5 Multiple Integrals Short Sheet (Calculus)

5.1 Development of 1-dim Reimann Integral-Line Integral

5.1.1 Intrduction (ds)

Introduction

$I = \int_C f(x,y)\,ds,\ C : r = r(\theta),\ \theta \in [\theta_1, \theta_2]$

$C : \begin{cases} x = r(\theta)\cos\theta \\ y = r(\theta)\sin\theta \end{cases},\ f(x,y) = f(r\cos\theta, r\sin\theta)$

$ds = \sqrt{x'(\theta)^2 + y'(\theta)^2}d\theta = \sqrt{r'^2 + r^2}d\theta$

$I = \int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r'^2 + r^2}\,d\theta$

5.1.2 I-Line Integral

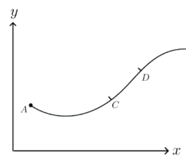
Definition

$I = \int_C f(x,y)\,ds = \left[\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}\right] = \int_{D_t} f(\varphi(t), \psi(t))\sqrt{\varphi'(t)^2 + \psi'(t)^2}\,dt$

A mean value property

$\rho(x,y) = f(x,y),\ \int_{C(A,B)} f(x,y)\,ds = M,\ \int_{C(C,D)} f(x,y)\,ds = m$

$\frac{\int_{C(C,D)} f(x,y)\,ds}{\int_{C(A,B)} f(x,y)\,ds} = \frac{\int_{C(C,D)} C\,ds}{\int_{C(A,B)} C\,ds} = \frac{S_{CD}}{S_{AB}}$



5.1.3 II-Line Integral

Definition

$I = \int_C f(x,y)\,dx = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t))\varphi'(t)\,dt$

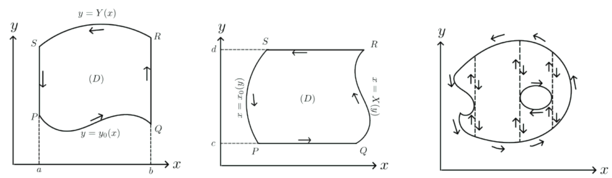
$I = \int_C f(x,y)\,dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} f(\varphi(t), \psi(t))\psi'(t)\,dt$

$I = \int_C P\,dx + Q\,dy = [x = \varphi(t), y = \psi(t)] = \int_{D_t} [P\varphi' + Q\psi']\,dt$

5.1.4 Calculate area using II-Line Integral

Calculate

$D = -\int_C y\,dx = \int_C x\,dy = \frac{1}{2}\int_C x\,dy - y\,dx$, where C is a simple closed cure with positive direction.



5.1.5 Conditions for curve integrals to be independent of the path

Theorem

In simply connected space D and given a path $C : A \to B$, II-line integral $\int_C P\,dx + Q\,dy$ is dependent of the integration path $C : A \to B$ i.f.f. $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Proof idea

Necessity "$\Rightarrow$":

By studying the differentiation method (partial differentiation with respect to x and y) of $F(x,y) = \int_{(x_0,y_0)}^{(x,y)} Pdx + Qdy$ and the primitive function of $Pdx + Qdy$ (regarding it as the total differential of some function) (i.e., $\exists\Phi$ such that $P = \frac{\partial \Phi}{\partial x}$, $Q = \frac{\partial \Phi}{\partial y}$ and $\int_C Pdx + Qdy = \Phi(x,y)\Big|_{(x_A,y_A)}^{(x_B,y_B)}$), we can conclude that the necessary and sufficient condition for $\int_C Pdx + Qdy$ to be path-independent is that $Pdx + Qdy$ $(= d\Phi(x))$ is an exact differential. The latter implies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ (necessity).

Sufficiency "$\Leftarrow$":

Fix $y \in [c,d]$, integrate $P = \frac{\partial \Phi}{\partial x}$ on $[x_0, x] \subset D$, then $\Phi(x,y) = \int_{x_0}^{x} Pdx + \Phi(x_0, y)$. Also, fix $x = x_0$, integrate $Q = \frac{\partial \Phi}{\partial y}$ on $[y_0, y] \subset D$, then $\Phi(x_0, y) = \int_{y_0}^{y} Q(x_0, y)dy + \Phi(x_0, y_0)$.

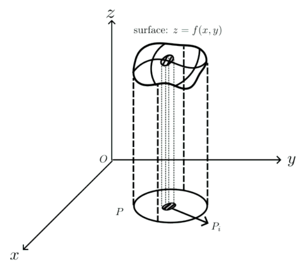
Then $\Phi(x,y) = \int_{x_0}^{x} P(x,y)dx + \int_{y_0}^{y} Q(x_0,y)dy + \Phi(x_0,y_0)$ or $\Phi(x,y) = \int_{y_0}^{y} Q(x,y)dy + \int_{x_0}^{x} P(x,y_0)dx + \Phi(x_0,y_0)$, which satisfies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

5.2 2-dim Multiple Integral

5.2.1 Definition

Definition

$\iint_P f(x,y)dP = \lim\limits_{\lambda=\max_i\{P_i\}\to 0} \sum\limits_i f(\xi_i,\eta_i)\Delta P_i = V$



5.2.2 Calculate: 2-dim Multiple Integral → Iterated Integral

2-dim Multiple Integral → Iterated Integral

Rectangle $P = [a,b] \times [c,d]$:

If $\iint_P f(x,y)dP$ exists and $I_1 = \int_a^b f(x,y)dx$ exists $\forall y \in [c,d]$, then $\iint_P f(x,y)dP = \int_c^d dy \int_a^b f(x,y)dx$.

If $\iint_P f(x,y)dP$ exists and $I_2 = \int_c^d f(x,y)dy$ exists $\forall x \in [a,b]$, then $\iint_P f(x,y)dP = \int_a^b dx \int_c^d f(x,y)dy$.

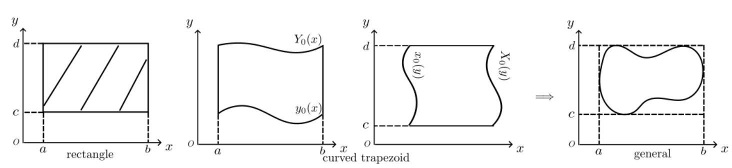
$\Longrightarrow$ If I_1 and I_2 exist, then $\iint_P f(x,y)dP = \int_a^b dx \int_c^d f(x,y)dy = \int_c^d dy \int_a^b f(x,y)dx$.

Curved trapezoid $P = [a,b] \times [y_0(x), Y(x)]$ or $P = [x_0(y), X(y)] \times [c,d]$:

If $\iint_P f(x,y)dP$ exists and $I_2 = \int_{y_0(x)}^{Y(x)} f(x,y)dy$ exists $\forall x \in [a,b]$, then $\iint_P f(x,y)dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x,y)dy$.

If $\iint_P f(x,y)dP$ exists and $I_1 = \int_{x_0(y)}^{X(y)} f(x,y)dx$ exists $\forall y \in [c,d]$, then $\iint_P f(x,y)dP = \int_c^d dy \int_{x_0(y)}^{X(y)} f(x,y)dx$.

$\Longrightarrow$ If I_1 and I_2 exist, then $\iint_P f(x,y)dP = \int_a^b dx \int_{y_0(x)}^{Y(x)} f(x,y)dy = \int_{y_0(x)}^{Y(x)} dy \int_a^b f(x,y)dx$.



5.2.3 Green's Formula: 2-dim Multiple Integral $\longleftrightarrow$ II-Line Integral

Green's Formula

$$\iint_D (\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y})\, dxdy = \oint_{C=\partial D} P\, dy + Q\, dx$$

or

$$\iint_D (\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y})\, dxdy = \oint_{C=\partial D} P\, dy - Q\, dx$$

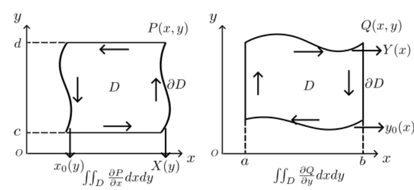
where D is a simply connected domain in $\mathbb{R}^2$ with boundary $C = \partial D$ and P, Q are continuously differentiable functions on D.

Proof idea

$P, Q, \frac{\partial P}{\partial x}, \frac{\partial Q}{\partial y}$ continuously on D, $\partial D \triangleq C$.

$\iint_D \frac{\partial P}{\partial x} dxdy = \int_c^d dy \int_{x_0(y)}^{X(y)} \frac{\partial P}{\partial x} dx = \int_c^d [P(X(y), y) - P(x_0(y), y)] dy = \oint_{\partial D} P(x,y) dy$

$\iint_D \frac{\partial Q}{\partial y} dxdy = \int_a^b dx \int_{y_0(x)}^{Y(x)} \frac{\partial Q}{\partial y} dy = \int_a^b [Q(x, Y(x)) - Q(x, y_0(x))] dx = -\oint_{\partial D} Q(x,y) dx$



5.2.4 Variable Substitution in Double Integrals

Intro

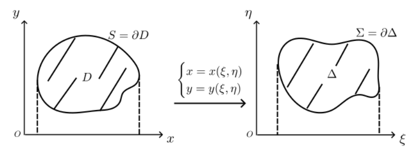
$D = \oint_{S=\partial D} xdy = \left[\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases} \right] = \oint_{\Sigma = \partial \Delta} x(\xi,\eta) \frac{\partial y}{\partial \xi} d\xi + x(\xi,\eta) \frac{\partial y}{\partial \eta} d\eta = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$

Analysis

As we could see, the essential of $\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}$ actually is coordinate transform between the x, y-plane and the ξ, η-plane.

$D = \oint_S xdy = \left[\begin{cases} x = x(\xi, \eta) \\ y = y(\xi, \eta) \end{cases}, \begin{cases} \xi = \xi(t) \\ \eta = \eta(t) \end{cases} \right] = \int_\alpha^\beta x(\xi(t), \eta(t)) (\frac{\partial y}{\partial \xi} \xi'(t) + \frac{\partial y}{\partial \eta} \eta'(t)) dt = \oint_\Sigma x(\xi,\eta) \frac{\partial y}{\partial \xi} d\xi + x(\xi,\eta) \frac{\partial y}{\partial \eta} d\eta$, let $x(\xi,\eta) \frac{\partial y}{\partial \xi} \triangleq P(x,y)$, $x(\xi,\eta) \frac{\partial y}{\partial \eta} \triangleq Q(x,y)$, then $D = \oint_\Sigma P d\xi + Q d\eta = (\frac{\partial P}{\partial \eta} = \frac{\partial Q}{\partial \xi}$, using Greeni's formula$) = \iint_\Delta (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta}) d\xi d\eta = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$.

Figures for aforementioned analysis



Idea of Analysis:

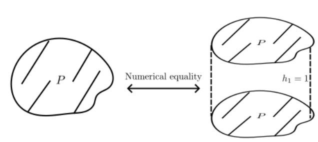
Using $\begin{cases} x = x(\xi,\eta) = x(\xi(t),\eta(t)) \\ y = y(\xi,\eta) = y(\xi(t),\eta(t)) \end{cases}$ (coordinate transform with parameterization), transform the II-line integration $\oint_S xdy$ (or $-\oint_S ydx$, or $\frac{1}{2}\oint_S xdy - ydx$) on x, y-plane to the II-line integral $\oint_\Sigma P(\xi,\eta)d\xi + Q(\xi,\eta)d\eta$, then using Green's formula to transform it to double integral $\iint_\Delta (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi d\eta$ on ξ, η-plane, following is how we calculate the double integral (often transform it to iterated integral $\int_{D\xi} d\xi \int_{\eta_0(\xi)}^{\eta(\xi)} (\frac{\partial Q}{\partial \xi} - \frac{\partial P}{\partial \eta})d\xi$, or others).

/Shortly, calculate the area of D on x, y plane: coordinate transform with parameterization $\rightarrow$ Green's formula $\rightarrow$ Double integration calculate/

Comparison:

Recaping the previous $D = \frac{1}{2}\oint_S xdy - ydx$, the original calculation method for this II-line integral is that find thier parametric equation $\begin{cases} x = x(t) \\ y = y(t) \end{cases}, t \in [a,b]$ for different (closed) curve S, then substitute into the calculation.From here, we could see that the essense of II-line integral is definite integral and has chirality.

However, "coordinate transform + Green's formula" method: $D = \iint_\Delta \frac{D(x,y)}{D(\xi,\eta)} d\xi d\eta$ as long as we master the coordinate transformation of a type of shape equation, we can easily calculate the results. Although there is a parameterization in the derivation, it is not reflected in the result (it can be seen from the process that if the parameterization is continued, it will be the result of the II-line integral in ξ, η-plane). In essence, the area is calculated as the volume with the same value, similiar with $\frac{1}{2}\oint_S xdy - ydx = \iint_D 1dxdy$ or $\oint_S xdy = \iint_D 1dxdy$ (If you unfold the left and right sides under the curved trapezoid, you can find the consistency.), it's just that the physical meanings of the two equations are different.



/Note: The choice of which method should be based on which is more convenient, parametric equations or coordinate transformations./

Variable Substitution in Double Integral

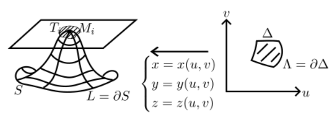
$$\iint_D f(x,y)dxdy = \begin{cases} x = x(\xi,\eta) \\ y = y(\xi,\eta) \end{cases} = \iint_\Delta f(x(\xi,\eta), y(\xi,\eta)) \left|\frac{D(x,y)}{D(\xi,\eta)}\right| d\xi d\eta$$

5.3 Surface Integral

5.3.1 Definition and basics

Definition

$S = \lim\limits_{\lambda\to 0} \sum\limits_i T_i$, where T_i is the projection area of S_i on the tangent plane at point $M_i \in S_i$.



Calculte Surface Integral(Omit matrix-vector operations)

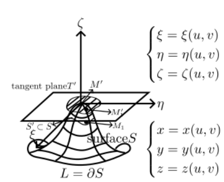
$$S = \lim_{\lambda\to 0} \sum_i T_i = \iint_\Delta \sqrt{A^2+B^2+C^2}dudv = \iint_\Delta \sqrt{EG-F^2}dudv$$

where $A = \begin{vmatrix} y'_u & z'_u \\ y'_v & z'_v \end{vmatrix}$, $B = \begin{vmatrix} z'_u & x'_u \\ z'_v & x'_v \end{vmatrix}$, $C = \begin{vmatrix} x'_u & y'_u \\ x'_v & y'_v \end{vmatrix}$, $E = \sum\limits_{x,y,z} x'^2_u$, $F = \sum\limits_{x,y,z} x'_u x'_v$, $G = \sum\limits_{x,y,z} x'^2_v$.

Procedure:

$\forall M_1 \in S'$, the coordinates of its projection M'_1 on T' are $\begin{cases} \xi = \xi(u,v) \\ \eta = \eta(u,v) \\ \zeta = \zeta(u,v) \end{cases}$,

then $S'_T = \iint_\Delta |\frac{D(\xi,\eta)}{D(u,v)}|dudv$, then the results could appear after a series of matrix-vector operations.



5.3.2 Surface Integral

I-Surface Integral

$$
\begin{aligned}
I &= \iint_S f(x,y,z)\,ds \\
&= [x = x(u,v), y = y(u,v), z = z(u,v), /parameterization/ \\
&\quad dS = \sqrt{A^2+B^2+C^2}\,du\,dv = \sqrt{EG-F^2}\,du\,dv] \\
&= \iint_\Delta f(x(u,v),y(u,v),z(u,v))\sqrt{A^2+B^2+C^2}\,du\,dv \\
&= \iint_\Delta f(x(u,v),y(u,v),z(u,v))\sqrt{EG-F^2}\,du\,dv
\end{aligned}
$$

Special case:
If $u = x,\ v = y,\ z = z(x,y)$,

$$
\begin{aligned}
I &= \iint_\Delta f(x,y,z(x,y))\sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy \\
&= \iint_\Delta f(x,y,z(x,y))\frac{dx\,dy}{|\cos(v)|}
\end{aligned}
$$

II-Surface Integral

$$
\begin{aligned}
I &= \iint_S f(x,y,z)\,dx\,dy = \text{or} \iint_S f(x,y,z)\,dy\,dz = \text{or} \iint_S f(x,y,z)\,dz\,dx \\
&\text{or} \iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy
\end{aligned}
$$

Connection between I and II-Surface Integrals

$$
\underbrace{\iint_S f(x,y,z)\,dx\,dy}_{(II)} = \underbrace{\iint_S f(x,y,z)\cos(v)\,ds}_{(I)}
$$

$$
\begin{aligned}
&\text{(Calculate it: parameterize to get double integral)} \\
&= [x = x(u,v), y = y(u,v), z = z(u,v), /parameterize/ \\
\cos(v) &= \pm\frac{C}{\sqrt{A^2+B^2+C^2}},\ dS = \sqrt{A^2+B^2+C^2}\,du\,dv] \\
&= \pm\iint_\Delta f(x(u,v),y(u,v),z(u,v))\cdot C\,du\,dv
\end{aligned}
$$

or

$$\underset{(II)}{\iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy} = \underset{(I)}{\iint_S P\cos(\lambda) + Q\cos(\mu) + R\cos(\nu)\,dS}$$

(Calculate it: parameterize to get double integral)

$$= [\begin{cases} x = x(u,v) \\ y = y(u,v) \quad /parameterize/ \\ z = z(u,v) \end{cases}$$

$$\cos(\lambda) = \pm\frac{A}{\sqrt{A^2+B^2+C^2}}, \cdots,$$

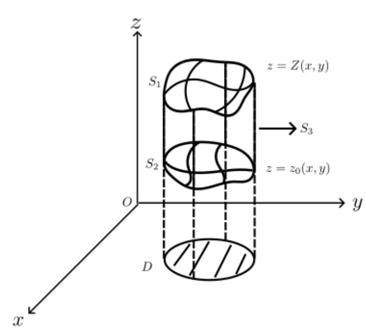
$$dS = \sqrt{A^2+B^2+C^2}\,du\,dv]$$

$$= \pm\iint_\Delta [P\cdot A + Q\cdot B + R\cdot C]\,du\,dv$$

5.3.3 Calculate Volume using Surface Integral

Calculate Volume using Surface Integral

$$\begin{aligned}
V &= \iint_D Z(x,y)\,dx\,dy - \iint_D z_0(x,y)\,dx\,dy \\
&= \iint_{S_2} z\,dx\,dy + \iint_{S_1} z\,dx\,dy \\
&= \iint_S z\,dx\,dy\ (S = S_1 + S_2 + S_3,\ \text{outside}) \\
or &= \iint_S x\,dy\,dz \\
or &= \iint_S y\,dz\,dx \\
or &= \frac{1}{3}\iint_S (x\,dy\,dz + y\,dz\,dx + z\,dx\,dy)
\end{aligned}$$



5.3.4 Stoke's Formula: II-Surface Integral $\longleftrightarrow$ II-Line Integral

Stoke's Formula $\Leftarrow$ Development of Green's Formula

$\int_{L=\partial S} P\,dx + Q\,dy + R\,dz = \iint_S (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\,dx\,dy + (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z})\,dy\,dz + (\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x})\,dz\,dx$

Proof idea

Using Green's formula on LHS as parameterizing $[x = x(u,v), y = y(u,v), z = z(u,v)]$, it's easy to prove.

5.4 3-dim Multiple Integral

5.4.1 Definition and basics

Definition

$\iiint_V f(x,y,z)\,dV = \lim_{\lambda \to 0} \sum_i f(x_i, y_i, z_i)\Delta V_i$.

Calculate

$$\begin{aligned}
\iiint_V f(x,y,z)\,dV &= \int_a^b dx \iint_{P_yz} f(x,y,z)\,dy\,dz \\
&(\iint_{P_yz} f(x,y,z)\,dy\,dz\ \exists) \\
&= \int_a^b dx \int_c^d dy \int_e^f f(x,y,z)\,dz \\
&(\int_e^f f(x,y,z)\,dz\ \exists) \\
\text{also} &= \iint_{P_xy} dx\,dy \int_e^f f(x,y,z)\,dz
\end{aligned}$$

$$\begin{aligned}
\iiint_V f(x,y,z)\,dV &= \int_{x_0}^X dx \iint_{P_yz} f(x,y,z)\,dy\,dz \\
&(\iint_{P_yz} f(x,y,z)\,dy\,dz\ \exists) \\
&= \int_{x_0}^X dx \int_{y_0}^{Y(x)} dy \int_{z_0(x,y)}^{Z(x,y)} f(x,y,z)\,dz \\
&(\int_{z_0(x,y)}^{Z_(x,y)} f(x,y,z)\,dz\ \exists) \\
\text{also} &= \iint_{P_xy} dx\,dy \int_{z_0(x,y)}^{Z_(x,y)} f(x,y,z)\,dz
\end{aligned}$$

5.4.2 Gauss's Formual 3-dim Multiple Integral $\longleftrightarrow$ II-Surface Integral

Gauss's Formula: $\Leftarrow$ Dimension-increasing version of Stokes' formula

$$\begin{cases} \iiint_V \frac{\partial R}{\partial z}\,dx\,dy\,dz = \iint_{S=\partial V} R\,dx\,dy \\ \iiint_V \frac{\partial P}{\partial x}\,dy\,dz\,dx = \iint_{S=\partial V} P\,dy\,dz \\ \iiint_V \frac{\partial Q}{\partial y}\,dz\,dx\,dy = \iint_{S=\partial V} Q\,dz\,dx \end{cases}$$

$\Sigma\ :\ \Longrightarrow\quad \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz \ = \ \iint_{S=\partial V} P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy$ (S is alonge the outside)

5.4.3 Variable Substitution in 3-dim Multiple Integral

Intro

$$V = \iint_S z\,dx\,dy$$

(coordinate transform)

$$= \iint_E z \cdot \frac{D(x,y)}{D(u,v)}\,du\,dv$$

$$= \iint_E z \cdot \left[\frac{D(x,y)}{D(\xi,\eta)}\frac{D(\xi,\eta)}{D(u,v)} + \frac{D(x,y)}{D(\eta,\zeta)}\frac{D(\eta,\zeta)}{D(u,v)} + \frac{D(x,y)}{D(\zeta,\xi)}\frac{D(\zeta,\xi)}{D(u,v)}\right] du\,dv$$

(II-Surface Integral) (using Green's formula)

$$= \iint_\Sigma z\frac{D(x,y)}{D(\xi,\eta)}\,d\xi\,d\eta + z\frac{D(x,y)}{D(\eta,\zeta)}\,d\eta\,d\zeta + z\frac{D(x,y)}{D(\zeta,\xi)}\,d\zeta\,d\xi$$

$$= \iiint_\Delta \left(\frac{\partial}{\partial \zeta}\left[z\frac{D(x,y)}{D(\xi,\eta)}\right] + \frac{\partial}{\partial \xi}\left[z\frac{D(x,y)}{D(\eta,\zeta)}\right] + \frac{\partial}{\partial \eta}\left[z\frac{D(x,y)}{D(\zeta,\xi)}\right]\right) d\xi\,d\eta\,d\zeta$$

(rearrange)

$$= \iint_\Delta |\frac{D(x,y,z)}{D(\xi,\eta,\zeta)}|\,d\xi\,d\eta\,d\zeta$$

Variable Substitution in 3-dim Multiple Integral

$$\iiint_D f(x,y,z)\,dx\,dy\,dz$$

$$= \left[\begin{cases} x = x(\xi,\eta,\zeta) \\ y = y(\xi,\eta,\zeta) \quad\quad /parameterize/ \\ z = z(\xi,\eta,\zeta) \end{cases}\right]$$

$$= \iiint_\Delta f(x(\xi,\eta,\zeta), y(\xi,\eta,\zeta), z(\xi,\eta,\zeta)) \cdot |\frac{D(x,y,z)}{D(\xi,\eta,\zeta)}|\,d\xi\,d\eta\,d\zeta$$

5.5 n-dim Multiple Integral

5.5.1 Definition and basics

Definition

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n = \lim_{\lambda(P)\to 0} \sigma(f, P, \xi)$$

where I is an n-dimensional interval, P is a partition of I, $\lambda(P)$ is the measure of P, and $\sigma(f, P, \xi)$ is the Riemann sum.

Calculate

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n$$

(Special)

$$= \int_{a_1}^{b_1} dx^1 \int_{a_2}^{b_2} dx^2 \cdots \int_{a_n}^{b_n} f(x^1, \cdots, x^n)\, dx^n$$

(General)

$$= \int_{x_1}^{X_1} dx^1 \int_{x_2(x_1)}^{X_2(x_1)} dx^2 \cdots \int_{x_n(x_1,\cdots,x_{n-1})}^{X_n(x_1,\cdots,x_{n-1}} f(x^1, \cdots, x^n)\, dx^n$$

5.5.2 Variable Substitution in n-dim Multiple Integral

Variable Substitution in n-dim Multiple Integral

$$\int_I f(x^1, \cdots, x^n)\, dx^1 \cdots dx^n$$

$$= \left[\begin{cases} x^1 = x^1(\xi^1, \cdots, \xi^n) \\ x^2 = x^2(\xi^1, \cdots, \xi^n) \\ \cdots \\ x^n = x^n(\xi^1, \cdots, \xi^n) \end{cases} \quad /parameterize/ \right]$$

$$= \int_\Delta f(x^1(\xi^1, \cdots, \xi^n), \cdots, x^n(\xi^1, \cdots, \xi^n)) \left| \frac{D(x)}{D(\xi)} \right| d\xi^1 \cdots d\xi^n$$

where $\frac{D(x)}{D(\xi)} = \frac{D(x^1,\cdots,x^n)}{D(\xi^1,\cdots,\xi^n)}$ is the Jacobian determinant of the transformation.

6 Multiple Integrals

6.1 The Riemann Integral over an n-Dimensional Interval

6.1.1 Comprehensions

First, the definition of Riemann integral over n-dimention interval is simimiar (almost same) with 1-dimension condition ($\int_I f(x)dx := \lim\limits_{\lambda(P)\to 0} \sigma(f, P, \xi)$),

$$\int_I f(x^1, \cdots, x^n)dx^1 \cdots dx^n \text{ or } \int \cdots \int_I f(x^1, \cdots, x^2)dx^1 \cdots dx^n$$

Then, introduce the necessity condition of it ($f \in \mathcal{R}(I) \implies f$ is bounded on I). After that, book gives some properties (Lemmas) about the set of measure zero (same with 1-dimension condition), and developes the Cantor Theorem (about uniform continuity of continuous functions on compact set). At last, we're given 2 criterions (necessity and sufficiency) of the Riemann integral (Darboux Criterion, Lebesgue Criterion).

Example. Let $f : I \to \mathbb{R}$ is a continuous real-value function on $I \subset \mathbb{R}^{n-1}$, then it's graph is a set measure zero in n-dimension.

Lemma 6.1. *[Development of Cantor Theorem] If the relation $\omega(f; x) < w_0$ holds at each point of a compact set K for the function $f : K \to \mathbb{R}$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\omega(f; U_\delta(x)) < w_0 + \varepsilon$ for each point $x \in K$.*

Theorem 6.1. *(Lebesgue's criterion). $f \in \mathcal{R}(I) \Leftrightarrow (f$ is bounded on $I) \wedge (f$ is continuous almost everywhere on $I)$.*

Theorem 6.2. *(Darboux Criterion) $f \in (I) \Leftrightarrow (f$ is bounded on $I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup\limits_P s(f, P)$, $\overline{\mathcal{J}} = \inf\limits_P S(f, P)$.*

6.1.2 Problems Sets

Problem 17. a) Show that a set of measure zero has no interior points.

b) Show that not having interior points by no means guarantees that a set is of measure zero.

c) Construct a set having measure zero whose closure is the entire space $\mathbb{R}^n$.

d) A set $E \subset I$ is said to have content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \ldots, I_k$ such that $\sum_{i=1}^{k} |I_i| < \varepsilon$. Is every bounded set of measure zero a set of content zero?

e) Show that if a set $E \subset \mathbb{R}^n$ is the direct product $\mathbb{R} \times e$ of the line $\mathbb{R}$ and a set $e \subset \mathbb{R}^{n-1}$ of $(n-1)$-dimensional measure zero, then E is a set of n-dimensional measure zero.

Proof. a) [Proof by contradiction] If E has a point x_0 as an interior point, then there exists a ball $B(x_0, r)$ such that $B(x_0, r) \subset E$. Since the ball has positive measure, it follows that E cannot have measure zero.

b) [Counterexample] Consider the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$, which have no interior points in $\mathbb{R}$ but are not of measure zero.

c) /inspired by b)/ Consider the set $E = \mathbb{Q}^n$, the set of all points in $\mathbb{R}^n$ with rational coordinates. The closure of E is $\mathbb{R}^n$, and E has measure zero since it can be covered by countably many intervals whose total length can be made arbitrarily small.

d) [Principle] No, since the Lebesgue measure zero set can be covered by at most countable intervals (which means measure $\mu = o(1) \cdot \mathcal{O}(1)$), while a set of content zero can be covered by a finite number of intervals (which means the content (also a kind of measure) sometimes is $\mathcal{O}(1) \cdot \mathcal{O}(1)$). [Example] The $\mathbb{Q} \cap [0, 1]$ and Cantor set are sets of measure zero but none of content zero.

e) From the construction of the product measure λ_n (Lebesgue measure of $\mathbb{R}^n$) /Lemma1/ if $A \in B(\mathbb{R}^k)$ and $B \in B(\mathbb{R}^{n-k})$, $(0 < k < n)$, then $\lambda_n(A \times B) = \lambda_k(A)\lambda_{n-k}(B)$. So here (using also that /Lemma2/ if $E_m \uparrow E$, $\lim_{m\to+\infty} \lambda_n(E_m) = \lambda_n(E)$) $\lambda_n(E) = \lambda_n(\mathbb{R}\times e)$ $= \lim_{m\to+\infty} \lambda_n((-m, m) \times e) = \lim_{m\to+\infty} \lambda_1((-m, m)) \times \lambda_{n-1}(e)$ $= \lim_{m\to+\infty} \lambda_1((-m, m)) \times 0 = 0$.

Notes. 1. The proof of e) /Lemma2/ could found in *Real and Complex Analysis* by Walter Rudin, Theorem 1.19(Page 16), based on the (positive) measure definition (countable additive of disjoint countable measurable sets).

Problem 18. a) Construct the analogue of the Dirichlet function in $\mathbb{R}^n$ and show that a bounded function $f : I \to \mathbb{R}$ equal to zero at almost every point of the interval I may still fail to belong to $\mathcal{R}(I)$.

b) Show that if $f \in \mathcal{R}(I)$ and $f(x) = 0$, at almost all points of the interval I, then $\int\limits_I f(x)\,\mathrm{d}x = 0$.

> *Proof.* a) [Counterexample] Consider $\mathcal{D}^n = \begin{cases} 1 & \text{if } x \in \mathbb{Q}^n \\ 0 & \text{if } x \in \mathbb{R}^n \backslash \mathbb{Q}^n \end{cases}$, the discontinuous set is $\mathcal{R}^n$.
>
> b) /by definition/ $\int_I f(x)dx = \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_i \in P} f(\xi_i)\Delta(x_i) = \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_i \in A\cap P} f(\xi_i)\Delta(x_i) + \lim\limits_{\lambda(P)\to 0} \sum\limits_{\xi_j \in B\cap P} f(\xi_j)\Delta(x_i) = \mathcal{O}(1) \cdot o(1) + o(1) \cdot \mathcal{O}(1)$, where $A = \{\cup[x_k, x_{k+1}] \subset P : \forall k, \exists \xi \in [x_k, x_{k+1}],\ s.t.\ f(\xi) \neq 0\}$ and $B = I \backslash A$.

Problem 19. There is a small difference between our earlier definition of the Riemann integral on a closed interval $I \subset \mathbb{R}$ and Definition 7 for the integral over an interval of arbitrary dimension. This difference involves the definition of a partition and the measure of an interval of the partition. Clarify this nuance for yourself and verify that

$$\int_a^b f(x)dx = \begin{cases} \int_I f(x)dx & \text{if } a < b \\ -\int_I f(x)dx & \text{if } a > b \end{cases}$$

where I is the interval on the real line $\mathbb{R}$ with endpoints a and b.

> *Proof.* Just understand it, it's not hard.

Problem 20. a) Prove that a real-valued function $f : I \to \mathbb{R}$ defined on an interval $I \subset \mathbb{R}^n$ is integrable over that interval if and only if for every $\varepsilon > 0$ there exists a partition P of I such that $S(f; P) - s(f; P) < \varepsilon$.

b) Using the result of a) and assuming that we are dealing with a real-valued function $f : I \to \mathbb{R}$, one can simplify slightly the proof of the sufficiency of the Lebesgue criterion. Try to carry out this simplification by yourself.

> *Proof.* a) /by The Darboux Criterion/ $f \in (I) \Leftrightarrow (f \textit{ is bounded on } I) \wedge (\underline{\mathcal{J}} = \overline{\mathcal{J}})$, where $\underline{\mathcal{J}} = \sup\limits_P s(f, P)$, $\overline{\mathcal{J}} = \inf\limits_P S(f, P)$. Notice that $\forall \varepsilon > 0,\ \exists P,\ s.t.\ S(f, P) - s(f, P) < \varepsilon \Leftrightarrow \underline{\mathcal{J}} = \overline{\mathcal{J}}$ (proof is similiar with 1-dimention condition), which is what we need to prove.
>
> b) [Segmented estimation] Sufficiency: procedure is same with book, a) just simplify the last step.

6.2 The Integral over a Set

6.2.1 Comprehensions

This part is not very long and mainly revolves around the concept of "admissible set".
First, the book introduces the concept (i.e. definition) of "admissible set" and some (operational) properties of it. Then, using a characteristic function χ_E of set E, we're given the definition of the integral over a set E and the corresponding Lebesgue's Criterion (over "admissible set"). At last, the book also gives the Jordan measure (or volume) of a bounded set E (actually it's only well defined for "admissible set").

Definition 1. *An <u>admissible set</u> is a <u>bounded</u> set $E \subset \mathbb{R}^n$ s.t. ∂E is <u>measure zero</u> in sense of Lebesgue.*

Lemma 6.2. $\forall$ *sets* $E,\ E_1,\ E_2 \subset \mathbb{R}^n$,

a) ∂E *is the closed set on* $\mathbb{R}^n$.

b) $\partial(E_1 \cup E_2),\ \partial(E_1 \cap E_2),\ \partial(E_1 \backslash E_2) \subset \partial E_1 \cup \partial E_2$.

c) Finite union or intersection of admissible sets is still admissible set, the difference of admissible sets is still admissible set.

Notes. 1. Generally, for infinite admissible sets, Lemma 2.2 c) is not true.
2. The boundary of admissible set is compact on $\mathbb{R}^n$ (closed and bounded).

Definition 2. *The integral of* f *over* E *is given by* $\int_E f(x)dx := \int_{\Gamma \supset E} f\chi_E(x)dx$, *where* I *is any interval containing* E. *If the integral exists, we say* $f \in \mathcal{R}(E)$.
$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$ *is the characteristic function of set* E *and* $f_{\chi_E}(x) = \begin{cases} f(x) & x \in E \\ 0 & x \notin E \end{cases}$.

Notes. 1. $\chi_E(x)$ is interrupted only on the boundary ∂E. The points of discontinuity of $f_{\chi_E}(x)$ are either points of discontinuity of f on E, or the result of discontinuities of χ_E, in which case they lie on ∂E.
2. $\forall I_1,\ I_2 \supset E$, integrals $\int_{I_1} f_{\chi_E}(x)dx$ and $\int_{I_2} f_{\chi_E}(x)dx$ either both exist (then values are equal) or both fail to exist.

Theorem 6.3. *A function* $f : E \to \mathbb{R}$ *is integrable over an admissible set if and only if it is bounded and continuous at almost all points of* E.

Notes. 1. Compared with f, f_{χ_E} may have additional points of discontinuity only on the ∂E.

Definition 3. *The (Jordan) measure or content of a bounded set $E \subset \mathbb{R}^n$ is $\mu(E) := \int_E 1 \cdot dx = \int_{I \supset E} \chi_E(x)dx$, provided this Riemann integral exists.*

Notes. 1. Only admissible sets, are measurable in the sense of aformentioned definition.

Comprehensions for Definition 3(From Book.)
The geometric meaning of $\mu(E)$ (if E is admissible) could be seen in following way:

$$\mu(E) = \int_{I \supset E} \chi_E(x)dx = \underline{\int}_{I \supset E} \chi_E(x)dx = \overline{\int}_{I \supset E} \chi_E(x)dx$$

Due to Darboux Criterion and Theorem, $\mu(E)$ is the common limit as $\lambda(P) \to 0$ of the volumes of polyhedra inscribed in and circumscribed about E, in agreement with the accepted idea of the volume of simple solids $E \subset \mathbb{R}^n$.
Why the $\mu(E)$ is called Jordan measure?

Definition 4. *A set $E \subset \mathbb{R}^n$ is a set of measure zero in the sense of Jordan or a set of content zero if for every $\varepsilon > 0$ it can be covered by a finite system of intervals $I_1, \ldots, I_k$ s.t. $\sum_{i=1}^{k} |I_i| < \varepsilon$.*

Compared with measure zero in the sense of Lebesgue, a requirement that the covering be finite appears here, shrinking the class of sets of Lebesgue measure zero. For example, the set of rational points is a set of measure zero in the sense of Lebesgue, but not in the sense of Jordan.
In order for the least upper bound of the contents of polyhedra inscribed in a bounded set E to be the same as the greatest lower bound of the contents of polyhedra circumscribed about E (and to serve as the measure $\mu(E)$ or content of E), it is obviously necessary and sufficient that the boundary ∂E of E have measure 0 in the sense of Jordan. That is the motivation for the following definition.

Definition 5. *A set E is Jordan-measurable if it is bounded and its boundary ∂E has Jordan measure zero.*

Hence, the class of Jordan-measurable subsets is precisely the class of admissible sets. That's the reason the measure $\mu(E)$ defined earlier can be called (and is called) the Jordan measure of the (Jordan-measurable) set E.